

Statistics for Data Science, Homework #3

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Instructions

You should submit a single PDF file of the homework via Moodle, with the PDF rendered directly from a Quarto or RMarkdown source file and not converted from any other output format.

Assignment

You will be working on a gene expression data set of 79 patients with leukemia belonging to two subgroups: patients with a chromosomal translocation (1) and patients cytogenetically normal (-1). The data are provided in the attached `gene_expr.tsv` file, containing expression for 2000 genes and an additional column with patient labels. Your goal is to build and compare support vector machine classifiers for predicting the leukemia subgroup from gene expression data. Your report should read as a self-contained mini scientific paper, following the guidelines and the feedback provided on the previous homeworks.

In particular, you should address the following points.

Data exploration

Briefly inspect the dataset. Include only summaries and visualizations that inform your subsequent analysis choices.

Analysis plan

Before fitting any model, outline your analysis strategy. This should cover:

- Which SVM variants you will consider and which hyperparameters need tuning for each;
- How you plan to evaluate predictive performance;
- How you plan to tune the hyperparameters and compare the different models.

Note: keep this section concise; a few sentences per point are sufficient. The bulk of your report should be devoted to the actual analysis and discussion of results.

Model fitting and interpretation on the full set of genes

Fit and tune the SVM variants you selected in your analysis plan, then compare them on the basis of their predictive performance. Support your comparison with appropriate evidence and comment on the results.

Variance-based gene filtering

A common approach in gene expression analysis is to keep only the most variable genes for downstream analysis, since most genes have low expression or do not vary much across samples and mainly contribute to noise. Repeat the analysis of the previous section after restricting the predictors to the genes whose standard deviation is among the top 5%.

Pay attention to *how* and *where* in the workflow this filtering step is applied. Compare the results obtained on the filtered data with those obtained on the full gene set.

Conclusions

Summarize your findings across the two analyses. Discuss the limitations of your analysis and any caveats that should accompany your results.

Please note that the **maximum allowed number of pages is 10**.